

Clinic Appointment System: Test Execution & Results

Project: Programming 1 Summative Project

Group: 5

Testing Date: 2026-09-21

Overview

This document records the results of the system testing phase. Each test case from the Test Plan was executed in the terminal. The system's actual behaviour was recorded and compared with the expected behaviour to determine a Pass/Fail status. Proof of execution is provided via terminal screenshots.

1. Input Validation & Error Handling

Test ID	Scenario Description	Test Input	Expected Output	Actual Output	Status
TC-V01	Empty string trap on Menus	Press Enter (Empty)	Prints "Oops sorry menu can not be left empty" and loops.	System printed the exact error message and looped back to the prompt.	PASS
TC-V02	Invalid data type on Integer inputs	"two"	Caught by ValueError. Prints "Invalid format" and loops.	ValueError was caught safely. System printed "Invalid format" and looped.	PASS
TC-V04	Date validation (Past Date)	"2026-09-20"	Prints "Appointment date cannot be before today's date" and loops.	System successfully blocked the past date and asked for a new input.	PASS

Evidence of Input Validation (TC-V01 & TC-V02):

```
=====
Clinic Appointment And Management System
=====

Welcome!

1. Login
2. Create Account(Patients Only)
3. Exit
select an option(1-3):
Oops sorry menu can not be left empty
select an option(1-3):
```

```
=====
Clinic Appointment And Management System
=====

Welcome!

1. Login
2. Create Account(Patients Only)
3. Exit
select an option(1-3): two
Invalid format. Select a number from 1-3
select an option(1-3):
```

```

---Book appointment---
Available Doctors
1. Dr. Smith
   Cardiology
Working hours: 09:00 - 17:00
select a doctor number: 1
You selected: Dr. Smith
Please enter your appointment date this format date as, YYYY-MM-DD: 2026-09-20
Appointment date cannot be before today's date.
Enter another date(YYYY-MM-DD: 2028-09-21
Please enter your appointment date this format date as, YYYY-MM-DD: 2028-09-20
1.09:00 - 10:00
2.10:00 - 11:00
3.11:00 - 12:00
4.12:00 - 13:00
5.13:00 - 14:00
6.14:00 - 15:00
7.15:00 - 16:00
8.16:00 - 17:00

```

2. Business Logic

Test ID	Scenario Description	Test Input	Expected Output	Actual Output	Status
TC-B01	Generate a unique Patient ID	Patient Registration	Generates a random 8-digit ID starting with "P-".	Generated "P-15590235".	PASS
TC-B04	Calculate free time slots	Select Doctor for Date	Returns 1-hour slots based on the doctor's shift.	Returned slots matching the doctor's 09:00 - 17:00 shift.	PASS

Evidence of Business Logic (TC-B04):

```
=====
Clinic Appointment And Management System
=====
```

Welcome!

1. Login
2. Create Account(Patients Only)
3. Exit

select an option(1-3): 2

```
=====
Patient Self-Registration
=====
```

Please enter your first name: John Doe

Invalid format. Please enter a valid name

Please enter your first name: John

Please enter your last name: Doe

Please enter your date of birth with this format date as, YYYY-MM-DD: 2002-08-09

Enter your gender (e.g., Male/Female): Male

Enter your mobile phone number(without the country code): 555-2967

Phone number must contain only numbers

Enter your mobile phone number(without the country code): 55551967

Enter a valid email address: johndoe@gmail.com

Enter your physical address: 16 Maple St

Generating your patients id

Your patient id is P-15590235

Registration successful!

Your patient ID is P-15590235 and your PIN is 4444.

Please use these credentials to login.

```

---Book appointment---
Available Doctors
1. Dr. Smith
   Cardiology
Working hours: 09:00 - 17:00
select a doctor number: 1
You selected: Dr. Smith
Please enter your appointment date this format date as, YYYY-MM-DD: 2026-09-20
Appointment date cannot be before today's date.
Enter another date(YYYY-MM-DD: 2028-09-21
Please enter your appointment date this format date as, YYYY-MM-DD: 2028-09-20
1.09:00 - 10:00
2.10:00 - 11:00
3.11:00 - 12:00
4.12:00 - 13:00
5.13:00 - 14:00
6.14:00 - 15:00
7.15:00 - 16:00
8.16:00 - 17:00
No time slots available for this date.

```

3. Data Persistence

Test ID	Scenario Description	Test Input	Expected Output	Actual Output	Status
TC-D01	Boot with missing JSON files	Delete admins.json	FileNotFoundException is caught. Boots smoothly.	System caught the error, created an empty list, and booted without crashing.	PASS
TC-D02	Save mixed Data on Shutdown	Select "3" (Exit)	Objects convert to dicts, writes to JSON files.	Terminal displayed "System closed successfully". JSON files populated.	PASS

Evidence of Data Persistence (TC-D02):

```
PS C:\Users\DELL\Group5_Programming1_Summative> & C:\Python314\python.exe c

=====
Clinic Appointment And Management System
=====

Welcome!

1. Login
2. Create Account(Patients Only)
3. Exit
select an option(1-3): 1
Enter your ID: A-ADMIN
Administrator Login is not connected yet.
Enter your 4-digit PIN: 1234

Opening Administrator Menu...

=====
Administrator Menu
=====

1.Register new patient
2.Display all patients
3.Search patient
4.Update patient information
5.Delete patient profile
6.Add new doctor
7.Display all doctors
8.Search doctor
9.Update doctor profile
10.Delete doctor profile
11.View all appointments
12.Cancel appointment
13. Logout
```

```
=====
Clinic Appointment And Management System
=====
```

```
Welcome!
```

- ```
1. Login
2. Create Account(Patients Only)
3. Exit
```

```
select an option(1-3): 3
```

```
Saving system data...
```

```
Thank you using our Clinic Management System.
```

```
System closed successfully!
```

```
PS C:\Users\DELL\Group5_Programming1_Summative> █
```

## 4. System Integration & Interface

| Test ID | Scenario Description      | Test Input         | Expected Output                       | Actual Output                                                     | Status |
|---------|---------------------------|--------------------|---------------------------------------|-------------------------------------------------------------------|--------|
| TC-I01  | Patient Self-Registration | "2", valid details | Creates patient, displays ID and PIN. | Patient successfully created and credentials displayed on screen. | PASS   |
| TC-I02  | Login Routing             | "DR-201", "5678"   | Routes to Doctor Menu.                | System recognized the DR- prefix and opened the Doctor Menu.      | PASS   |

|               |                                |                    |                                     |                                                         |             |
|---------------|--------------------------------|--------------------|-------------------------------------|---------------------------------------------------------|-------------|
| <b>TC-I03</b> | End-to-End Appointment Booking | Patient books slot | Appointment ID generated and saved. | Success message displayed with Appointment ID and time. | <b>PASS</b> |
|---------------|--------------------------------|--------------------|-------------------------------------|---------------------------------------------------------|-------------|

#### Evidence of Integration (TC-I01 & TC-I02):

```
=====
Clinic Appointment And Management System
=====

Welcome!

1. Login
2. Create Account(Patients Only)
3. Exit
select an option(1-3): 2

=====
Patient Self-Registration
=====

Please enter your first name: John Doe
Invalid format. Please enter a valid name
Please enter your first name: John
Please enter your last name: Doe
Please enter your date of birth with this format date as, YYYY-MM-DD: 2002-08-09
Enter your gender (e.g., Male/Female): Male
Enter your mobile phone number(without the country code): 555-2967
Phone number must contain only numbers
Enter your mobile phone number(without the country code): 55551967
Enter a valid email address: johndoe@gmail.com
Enter your physical address: 16 Mapel St
Generating your patients id
Your patient id is P-15590235

Registration successful!
Your patient ID is P-15590235 and your PIN is 4444.

Please use these credentials to login.
```

```
=====
Clinic Appointment And Management System
=====
```

Welcome!

```
1. Login
2. Create Account(Patients Only)
3. Exit
select an option(1-3): 1
Enter your ID: D-201
Invalid ID format. Please try again.
```

```
=====
Clinic Appointment And Management System
=====
```

Welcome!

```
1. Login
2. Create Account(Patients Only)
3. Exit
select an option(1-3): 1
Enter your ID: DR-201
Enter your 4 digit PIN: 5678
Login successful
```

Opening Doctor Menu...

```
=====
 Doctor Menu - Dr. Smith
=====
```

```
1.View My Schedule
2.View My profile
3.Update Appointment Status
4.Logout
Choose an option(1-4):
```

Opening Patient Menu...

```
=====
Patient Menu - John Doe
=====
```

```
1. View my information
2. Book appointment
3. View my appointment
4. Cancel appointment
5. Logout
Choose an option(1-5): 2
```

---Book appointment---

Available Doctors

```
1. Dr. Smith
 Cardiology
```

Working hours: 09:00 - 17:00

select a doctor number: 1

You selected: Dr. Smith

Please enter your appointment date this format date as, YYYY-MM-DD: 2026-10-02

```
1. 09:00 - 10:00
2. 10:00 - 11:00
3. 11:00 - 12:00
4. 12:00 - 13:00
5. 13:00 - 14:00
6. 14:00 - 15:00
7. 15:00 - 16:00
8. 16:00 - 17:00
```

Select a slot number: 2

Your appointment ID is APT-65352706

Success! Appointment booked.

Appointment ID: APT-65352706 | Date: 2026-10-02 | Time: 10:00 - 11:00